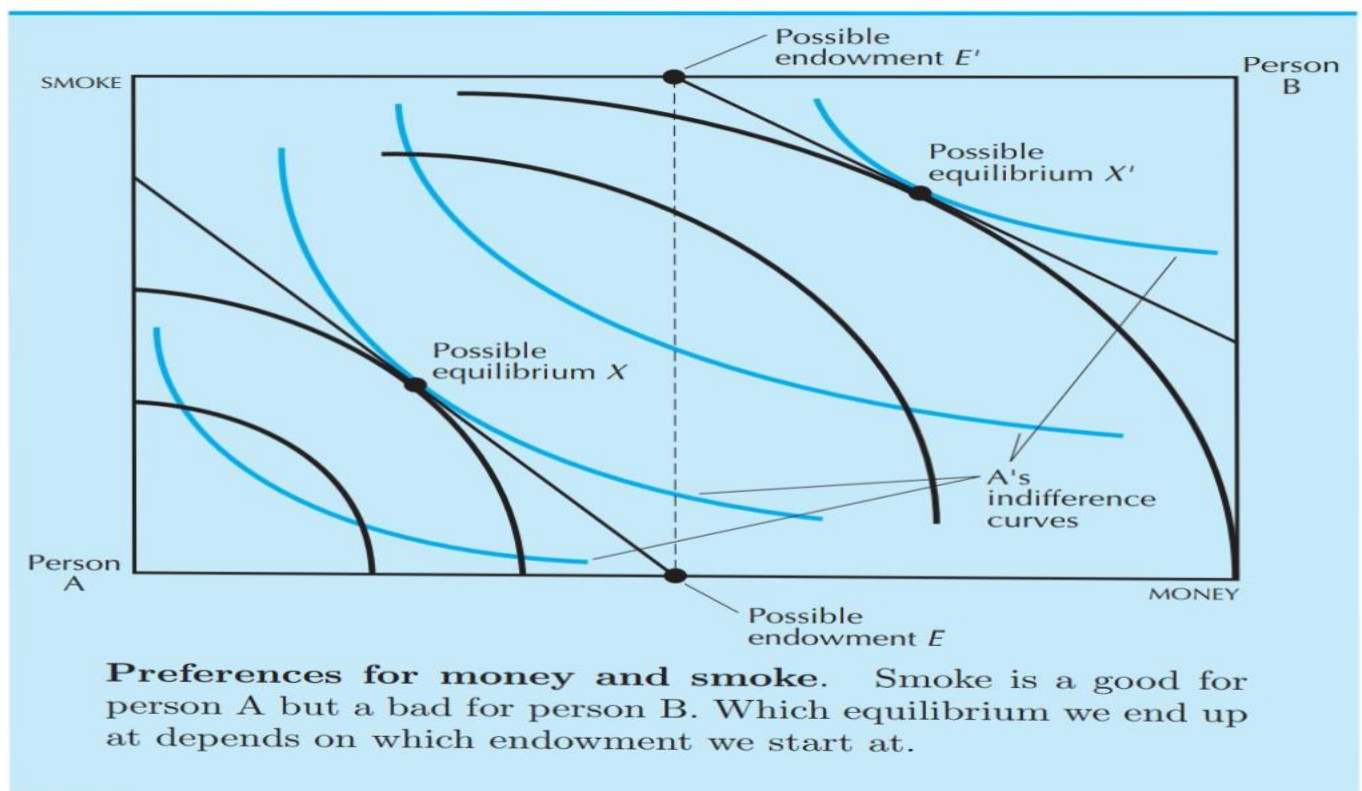


1(a) How a smoker and nonsmoker sharing the same room can ensure pareto optimum smoke in the room under two situation

a)everybody has the right to have fresh air b)everybody has the right to smoke

Show that the outcome depends on nature of property right and distribution of income.

Answer: We'll imagine two roommates, A and B, who have preferences over "money" and "smoke." We suppose that both consumers like money, but that A likes to smoke and B likes clean air. We depict the consumption possibilities for the two consumers in an Edgeworth box. The length of the horizontal axis will represent the total amount of money the two agents have, and the height of the vertical axis will represent the total amount of smoke that can be generated. The preferences of agent A are increasing in both money and smoke, while agent B's preferences are increasing in money and clean air—the absence of smoke. We'll measure smoke on a scale from 0 to 1, where 0 is no smoke at all, and 1 is the proverbial smoke-filled room. The amount of smoke is a good for A and a bad for B, so that B is moved to a more preferred position as A consumes less smoke. A's money horizontally from the lower left-hand corner of the box, and B's money horizontally from the upper right-hand corner. But the total amount of smoke is measured vertically from the lower left-hand corner.



In the ordinary Edgeworth box diagram B is made better off when A reduces his consumption of good 2—but that is because B then gets to consume more of good 2. In the Edgeworth box in Figure 1, B is also better off when A reduces his consumption of good 2 (smoke). In this example, B is better off when A reduces his consumption of smoke since both agents must consume the same amount of smoke and

smoke is a bad for agent B. Let's assume that they both have the same amount of money, say \$100 apiece, so that their endowments will lie somewhere on the vertical line EE' in Figure 1, in order to determine exactly where on this line the endowments lie, we must determine the initial "endowment" of smoke/clean air. It depends on the legal rights of smokers and nonsmokers. A has an initial endowment of \$100 means that A can decide to consume the \$100 himself, or he can give it away or trade it to any other individual. There is a legal definition of property involved in saying that a person "owns" or "has a right to" \$100. Similarly if a person has a property right to clean air, it means that he can consume clean air if he wants to, or he can give it away or sell that right to someone else. In this way, having a property right to clean air is no different from having a property right to \$100. considering a legal situation where person B has a legalright to clean air. Then the initial endowment in Figure 1 is labeled E; it is where A has (100, 0) and B has (100, 0). This means that both A and B have \$100, and that the initial endowment—what there would be in the absence of trade—is clean air. A pareto efficient allocation is one where neither consumer can be made better off without the other being made worse off. Such an allocation will be characterized by the usual tangency condition that the marginal rates of substitution between smoke and money should be the same between the two agents, as illustrated in Figure 1. It is easy to imagine A and B trading to such a Pareto efficient point. In effect, B has the right to clean air, but he can allow himself to be "bribed" to consume some of A's smoke. in the case with no externalities, there is no reason why the initial endowment is Pareto efficient. One of the aspects of having a property right to clean air is having the right to trade some of it away for other desirable goods—in this case, for money. It can easily happen that B would prefer to trade some of his right to clean air for some more money. The point labeled X in Figure 1 is an example of such a case.